Linear Algebra: Review

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Linear Algebra: Review
(Matrices)

Learning Objectives

In this module, we will study

- Matrix properties
- System of linear equations
- Matrix Factorization
  - Eigenvalues and eigenvectors
  - Eigenvalue Decomposition (EVD)
  - Singular Value Decomposition (SVD)
- Taylor Series

Matrix Properties

- **Matrix Operations**: transpose, trace, determinant, adjoint, algebraic operations on matrices
  (addition and multiplication)

- The **rank** of a matrix $\mathbf{A} \in \mathrm{R}^{m\times n}$, rank($\mathbf{A}$), is the size of the largest linearly independent set of columns of $\mathbf{A}$. Rank satisfies rank($\mathbf{A}$) $\leq \min\{m,n\}$.

  $A = 3\times 2$
  rank $(A) \leq 2$

  $A = 3\times 4$
  rank $(A) \leq 3$

- For matrices $\mathbf{A} \in \mathrm{R}^{\overset{3\times 2}{m\times k}}$ and $\mathbf{B} \in \mathrm{R}^{\overset{2\times 4}{k\times n}}$, we have

  $$\text{rank}(\mathbf{A})+\text{rank}(\mathbf{B})-k \leq \text{rank}(\underbrace{\mathbf{AB}}_{3\times 4}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}.$$

- For the two matrices of same size, we have

  $$\text{rank}(\mathbf{A} + \mathbf{B}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}).$$

- The **trace** of a matrix is the sum of its main diagonal entries, that is, trace($\mathbf{A}$) = a_{ii}

- For two matrices **A** and **B**, trace(**AB**) = trace(**BA**)

System of Linear Equations

- The **range** of a matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$, range($\mathbf{A}$), is the set

  range($\mathbf{A}$) = $\{\mathbf{b} \in R^m \mid \exists\ \mathbf{x} \in R^n$ with $\mathbf{Ax} = \mathbf{b}\}$

  Equivalently, the range of **A** is the set of all linear combinations of columns of **A**.

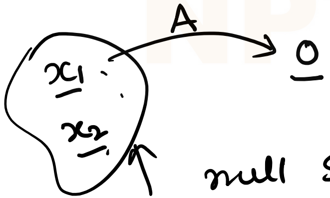
$$\underset{m\times n}{A}\ \underset{n\times 1}{\underline{x}} = \underset{m\times 1}{\underline{b}} \qquad A: \mathbb{R}^n \longrightarrow \mathbb{R}^m$$

$$\underline{x} \longrightarrow \underline{b}$$

$$\underset{3\times 3}{\begin{bmatrix} \underline{a_1} & \underline{a_2} & \underline{a_3} \\ \downarrow & \downarrow & \downarrow \end{bmatrix}} \underset{3\times 1}{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}} = \begin{bmatrix} \underline{b} \end{bmatrix}$$

$$= \underline{a_1} x_1 + \underline{a_2} x_2 + \underline{a_3} x_3 = \underline{b}$$

- The **nullspace** of a matrix, null($\mathbf{A}$) is the set of all vectors **x** such that $\mathbf{Ax} = \mathbf{0}$.



$$A\underline{x_1} = \underline{0}$$

$$A\underline{x_2} = \underline{0}$$

null space of A

System of Linear Equations

- The equation is **consistent** if there exists a solution **x** to this equation; equivalently, we have rank([**A**, **b**]) = rank(**A**) (Rouche–Capelli theorem), or **b** ∈ range(**A**).

$$A\underline{x} = \underline{b}$$

$$[A \mid \underline{b}]$$

- The equation has a **unique solution** if rank([**A**, **b**]) = rank(**A**) = n.
- The equation has **infinitely many solutions**

  if rank([**A**, **b**]) = rank(**A**) < n.
- The equation has **no solution** if rank([**A**, **b**]) > rank(**A**).

$$\overset{2}{\text{rank}([A, b])} > \overset{1}{\text{rank}(A)}$$

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \underline{x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \Rightarrow \text{has no soln.}$$

rank = 1

$$[(A \mid b)] = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

rank = 2

- A matrix **A** ∈ $R^{m\times n}$ is **nonsingular** if **Ax** = 0 if and only if **x**= 0.

Matrix Properties Continued

- A **normal matrix** is a matrix **N** such that $\mathbf{N}\mathbf{N}^H = \mathbf{N}^H\mathbf{N}$.
- A **Hermitian matrix** is one such that $\mathbf{A}^H = \mathbf{A}$. A real valued Hermitian matrix is called a **symmetric matrix**.
- A **skew Hermitian matrix** is one such that $\mathbf{A}^H = -\mathbf{A}$.
- A **unitary matrix** is a square matrix with $\mathbf{U}\mathbf{U}^H = \mathbf{I} = \mathbf{U}^H\mathbf{U}$. $\Rightarrow$ $U^H = U^{-1}$
- A real-valued unitary matrix is called an **orthogonal matrix**.
- An **idempotent matrix** satisfies $\mathbf{A}^2 = \mathbf{A}$.
- A **Projection matrix P** satisfies $\mathbf{P}^2 = \mathbf{P}$ (**P** is idempotent). If $\mathbf{P} = \mathbf{P}^H$, then **P** is called an **orthogonal projection**.

$$A = \begin{pmatrix} 1 & 2 & 3j \\ 4 & 5 & 0 \\ 2j & 0 & 1 \end{pmatrix} \qquad A^H = \begin{pmatrix} 1 & 4 & -2j \\ 2 & 5 & 0 \\ -3j & 0 & 1 \end{pmatrix}$$

Matrix Decomposition (Factorization)

Matrix factorization is a method of expressing a given matrix as a product of two or more matrices

$A = BCD$

- Need: to simplify the matrix by breaking it down into smaller, more manageable components; the properties of component matrices is helpful in many applications.

- Types
  - Eigenvalue decomposition (EVD)
  - Singular value decomposition (SVD)

Note: There are many other types of matrix decomposition, such as QR, Cholesky decomposition, NMF, and so on.

Eigenvalues and Eigenvectors

Definition: A matrix **K** is called positive semi-definite, if the quadratic form

$$\mathbf{z}^T\mathbf{K}\mathbf{z} \geq 0 \quad \forall \ \mathbf{z} \neq \mathbf{0},$$

and is positive definite (p.d.), if the quadratic form

$$\mathbf{z}^T\mathbf{K}\mathbf{z} > 0 \quad \forall \ \mathbf{z} \neq \mathbf{0}.$$

Definition: The **eigenvalues** of an *nxn* matrix **K** are those numbers λ for which the characteristic equation

$\mathbf{K}\boldsymbol{\phi}=\lambda\boldsymbol{\phi}$ has a solution for $\boldsymbol{\phi} \neq \mathbf{0}$,

where $\boldsymbol{\phi}$ = eigenvector and λ = eigenvalue.

Eigenvectors are normalized vectors having unit norm, i.e., $\boldsymbol{\phi}^T\boldsymbol{\phi} = \|\boldsymbol{\phi}\|^2 = 1$.

Theorem: λ is an eigenvalue of the square matrix **K** if and only if

$$\det(\mathbf{K}-\lambda\mathbf{I}) = 0.$$

$K_{n\times n}$ $\qquad z \neq 0$

$z_{n\times 1}$ $\qquad \underbrace{\underset{1\times n}{z^T}\ \underset{n\times n}{K}\ \underset{n\times 1}{z}}_{1\times 1} \geq 0$

$n=3$

$$\begin{matrix} 1 & 2 & 1 \\ 0 & 3 & 2 \\ 0 & 1 & 2 \end{matrix}$$

$\underbrace{\underset{n\times n}{K}\ \underset{n\times 1}{\phi}}_{n\times 1} = \lambda \underset{n\times 1}{\phi}$

$\det(K-\lambda I)=0$

$K\phi = \lambda\phi$

$\|\phi\| = \langle \phi, \phi \rangle^{1/2}$

$= \sqrt{\phi^T\phi}$

$= \sqrt{\sum_{i=1}^{n} \phi_i \phi_i}$

$= \sqrt{\sum_{i=1}^{n} \phi_i^2}$

Eigenvalues and Eigenvectors

Definition: Two *nxn* matrices are called similar if there exists an *nxn* matrix **T** with det(**T**) $\neq$ 0, s. t. $\mathbf{T^{-1}AT=B}$, where **A** and **B** are similar matrices.

Theorem: An *nxn* matrix **K** is similar to a diagonal matrix if and only if **K** has *n*-linearly independent eigenvectors and hence,

$$\mathbf{U^{-1}KU} = \Lambda \quad \Rightarrow \quad K = U \Lambda U^{-1} = U \Lambda U^T$$

$n \times n$ diagonal matrix

$$U^T U = U U^T = I$$

Theorem: Let **K** be a real symmetric matrix with eigenvalues $\lambda_1, \lambda_2, \ldots\lambda_n$, . Then, **K** has *n* mutually orthogonal eigenvectors $\phi_1, \phi_2, \ldots, \phi_n$.

Theorem: Let **K** be a real symmetric matrix with eigenvalues $\lambda_1, \lambda_2, \ldots\lambda_n$.
Then, **K** is similar to a diagonal matrix, i.e.,

$$\mathbf{U^{-1}KU} = \Lambda,$$

where the columns of U contain the ordered orthogonal unit eigenvectors $\phi_1, \phi_2, \ldots, \phi_n$ of **K**.

Eigenvalues and Eigenvectors

Theorem: A real symmetric matrix **K** is positive definite if and only if all its eigenvalues λ_1, λ_2, …..λ_n, are positive.

Note: Since det(**K**) is the product of its eigenvalues, det(**K**) > 0. This implies that the matrix **K** is full rank.

Let us Try!

Question: Perform eigenvalue decomposition on the following matrix.

$$\mathbf{A} = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

You may pause the video and try.

Let us Try!

Question: Perform eigenvalue decomposition on the following matrix.

$$\mathbf{A} = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

Answer:

$$\det(A - \lambda I) = 0$$

$$\det\left[\begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right] = 0$$

$$\det\begin{pmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{pmatrix}$$

$$= (5-\lambda)(2-\lambda) - 4 = 0$$

$$\Rightarrow \lambda = 1; \ \lambda = 6$$

$$A\underline{\phi} = \lambda \underline{\phi}$$

$$\begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} = \underline{1}\begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}$$

$$\Rightarrow 5\phi_1 + 4\phi_2 - \phi_1 = 0$$

$$\phi_1 + 2\phi_2 - \phi_2 = 0$$

$$4\phi_1 + 4\phi_2 = 0$$

$$\Rightarrow \phi_1 + \phi_2 = 0$$

$$\Rightarrow \phi_1 = -\phi_2$$

$$\underline{\phi_1} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$\Downarrow$ normalize

$$\underline{\phi_1} = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

Let us Try!

$$\Lambda = \begin{pmatrix} 1 & 0 \\ 0 & 6 \end{pmatrix}$$

capital lambda

$$U = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{4}{\sqrt{17}} \\ \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{17}} \end{pmatrix}$$

$$A = U \Lambda U^T$$

Try It Yourself!

Question: Perform eigenvalue decomposition on the following matrix.

$$\mathbf{A} = \begin{bmatrix} 2 & 7 & 8 \\ 5 & 7 & 1 \\ 0 & 4 & 3 \end{bmatrix}$$

This is for your own practice. Solution will not be provided for this problem.

Singular Value Decomposition (SVD)

Let $\mathbf{A} \in C^{m\times n}$. Then, a **singular value decomposition (SVD)** of **A** is given by

$$\underset{m\times n}{\mathbf{A}} = \underset{m\times m}{\mathbf{U}}\,\underset{m\times n}{\Sigma}\,\underset{n\times n}{\mathbf{V}^{H}}, \text{ where}$$

- $\mathbf{U} \in C^{m\times m}$ is unitary
- $\Sigma = diag(\sigma_1, \sigma_2, \cdots, \sigma_p) \in R^{m\times n}$ is diagonal with
- $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p \geq 0$ [$p = \min(m, n)$] and
- $\mathbf{V} \in C^{n\times n}$ is unitary

$A_{4\times 3}$ $\Sigma_{4\times 3}$

$A_{3\times 4}$ $\Sigma_{3\times 4}$

$$\begin{bmatrix} \sigma_1 & 0 & 0 & 0 \\ 0 & \sigma_2 & 0 & 0 \\ 0 & 0 & \sigma_3 & 0 \end{bmatrix} \qquad \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \\ 0 & 0 & 0 \end{bmatrix}$$

Singular values

The values $\sigma_i = \sigma_i(\mathbf{A})$ are called the **singular values** of **A** and are uniquely determined as the positive square roots of the eigenvalues of $\mathbf{A}^{H}\mathbf{A}$.

$\underset{3\times 4}{A}$

$$\underbrace{\underset{n\times m}{A^H}\;\underset{m\times n}{A}}_{\substack{n\times n \\ 4\times 4}} \qquad \underbrace{\underset{m\times n}{A}\;\underset{n\times m}{A^H}}_{\substack{m\times m \\ 3\times 3}}$$

Let us Try!

Question: Perform singular value decomposition on the following matrix.

$$\mathbf{A} = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$$

You may pause the video and try.

$$\underbrace{\underset{2\times 3}{A}\ \underset{3\times 2}{A^T}}_{2\times 2} \equiv \begin{pmatrix} 17 & 8 \\ 8 & 17 \end{pmatrix} \Rightarrow \begin{array}{l} \lambda_1 = 25 \\ \lambda_2 = 9 \end{array}$$

$$\boxed{\begin{array}{l} \sigma_1 = 5 \\ \sigma_2 = 3 \end{array}}$$

$$\underset{2\times 3}{A} = \underset{2\times 2}{U}\ \underset{2\times 3}{\Sigma}\ \underset{3\times 3}{V^T}$$

$$A^T A \Rightarrow \begin{array}{l} \lambda_1 = 25 \\ \lambda_2 = 9 \\ \lambda_3 = 0 \end{array}$$

Try It Yourself!

Question: Perform singular value decomposition on the following matrix.

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 3 \\ 3 & 1 & 1 \end{bmatrix}$$

This is for your own practice. Solution will not be provided for this problem.

Taylor Series

- A mathematical tool to represent a function as an infinite sum of terms, where each term is a derivative of the function evaluated at a specific point
- Used to approximate the value of a function
- The general form of a Taylor series for a function $f(x)$ centered at a point a is:

$$f(x) = f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \frac{f'''(a)(x-a)^3}{3!} + \cdots$$

where,

$f'(a)$: first derivative of $f(x)$ evaluated at x = a

$f''(a)$: second derivative of $f(x)$ evaluated at x = a

… and so on

!: factorial

Brook Taylor FRS
(Aug. 18, 1685 – Dec. 29, 1731)

https://galileo-unbound.blog/2020/08/03/brook-taylors-infinite-series/

Summary

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